Statistical Business & Customer Churn Analysis Report

This report presents a complete statistical analysis of business sales and customer churn datasets. The analysis includes descriptive statistics, correlation analysis, hypothesis testing, and regression analysis to generate actionable business insights.

1. Descriptive Statistics

Metric	Value
Average Sales	123650.48
Median Sales	97955.50
Sales Standard Deviation	100161.09
Average Monthly Charges	113.64
Average Customer Tenure	36.53

2. Correlation Analysis

- Quantity vs Total Sales Correlation: 0.69
- Price vs Total Sales Correlation: 0.65
- Tenure vs Total Charges Correlation: -0.01

Interpretation:
Strong positive correlation values indicate that as one variable increases, the related business metric also increases significantly.

3. Hypothesis Testing

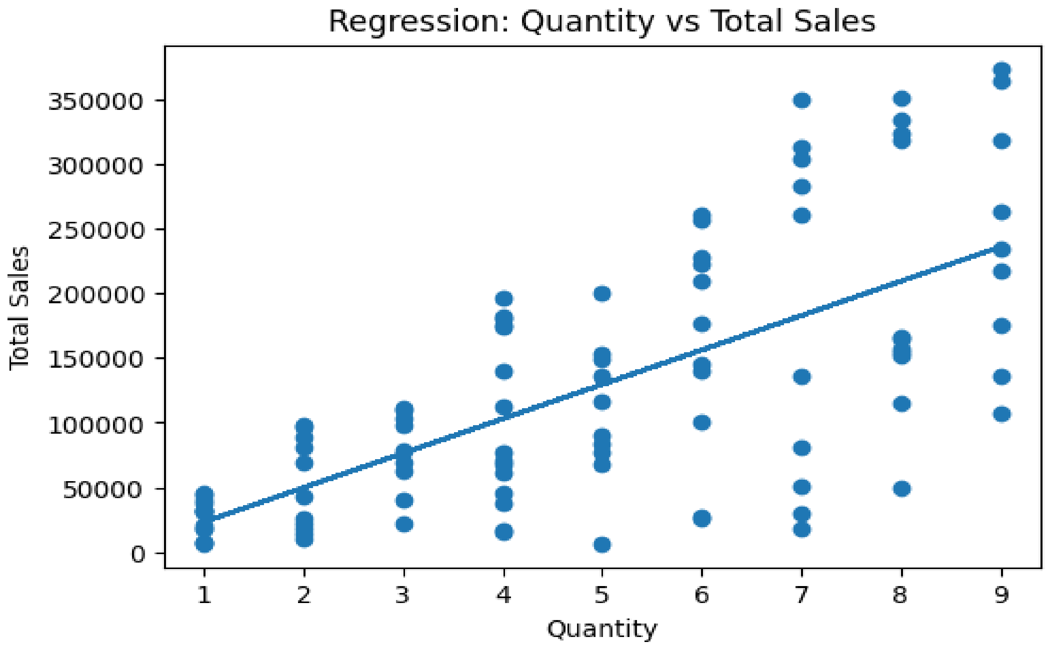
T-statistic: -0.3571
P-value: 0.7227

Result: No statistically significant difference was found between regional sales.

4. Regression Analysis

Intercept: -3638.74
Slope: 26629.54
R-squared: 0.47

Interpretation:
The regression model explains the relationship between product quantity and total sales. A higher R-squared value indicates better predictive performance.



5. Business Insights & Recommendations

1. Higher product quantity strongly contributes to increased sales revenue.
2. Customers with longer tenure tend to generate higher total charges.
3. Monthly charges influence customer churn behavior.
4. Statistical analysis can support future business forecasting and decision-making.
5. Businesses should focus on customer retention and high-performing products.

Conclusion

This project demonstrates how statistical techniques can be applied to real-world business datasets. The combination of descriptive statistics, correlation analysis, hypothesis testing, and regression analysis provides valuable insights that support data-driven business decisions.